

An Introduction to the Mobile Segment

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Mobile Solutions



HOME
SOLUTIONS



ENTERPRISE
SOLUTIONS



MOBILE
SOLUTIONS



EMBEDDED
SOLUTIONS

EMERGING APPLICATIONS

Battery Operated, On the Move

The mobile segment		TAM 2006*	TAM 2011*	Avg. SoCs per Product (2011)*
■ Smart Phone		100M	350M	3.0
■ Feature Phone		395M	420M	1.5
■ Voice Phone		405M	330M	1.0
■ Cordless Phone		130M	165M	1.0
■ Bluetooth (ex phone.)		110M	350M	1.0
■ Portable Media Players		150M	250M	1.0
		1,290M (1,685M)	1,865M (2,775M SoCs)	

*Mixture of ARM and industry analysts

1 Billion ARM Processors in 2005

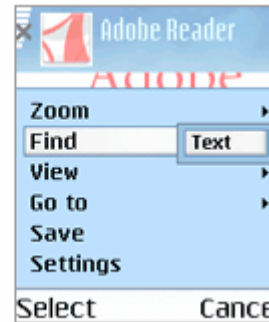


Trends: Operators Focus on Services



Needs high-performance platforms with enhanced security

Productivity and Value Added Services



ARM11™ family

250-500MHz

Office applications

PMP inc Video

Jazelle DBX with MVM

Great 3D OpenGL-ES

Energy Mgmt (IEM)

Security (TrustZone®)

\$25/month Dataplan &

\$15/month Services

ARM9™ family

100-250MHz

Push E-mail

Portable Media Player (Software
or AudioDE™)

Good Java experience (Jazelle®)

PS1 equiv gaming experience

(Swerve client)

Adobe Flashlite UI

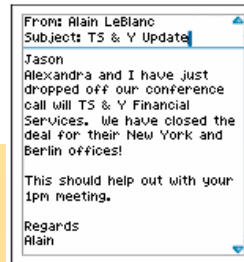
\$20/month data plan

ARM7™ family

<50MHz

Text & Voice

\$ 0.10 per text



Trends: OEMs Deliver Convergence

- Phones as watches
- Phones as MP3 players
- Phones as cameras
- Phones as gaming consoles
- Phones as video cameras/TV

But...

- Battery life limits use
- Ease of use is key
- Market for low cost, voice centric still strong
- Needs versatile processing power and a focus on low power



Gaming and New \$ Opportunities

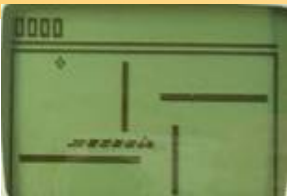


ARM11 family
250-500MHz
PMP inc Video
Jazelle + DAC
Great 3D OpenGL-ES
Energy Mgmt (IEM)
Security (TrustZone)
\$10 per game



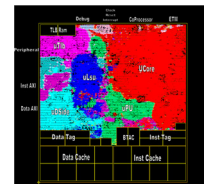
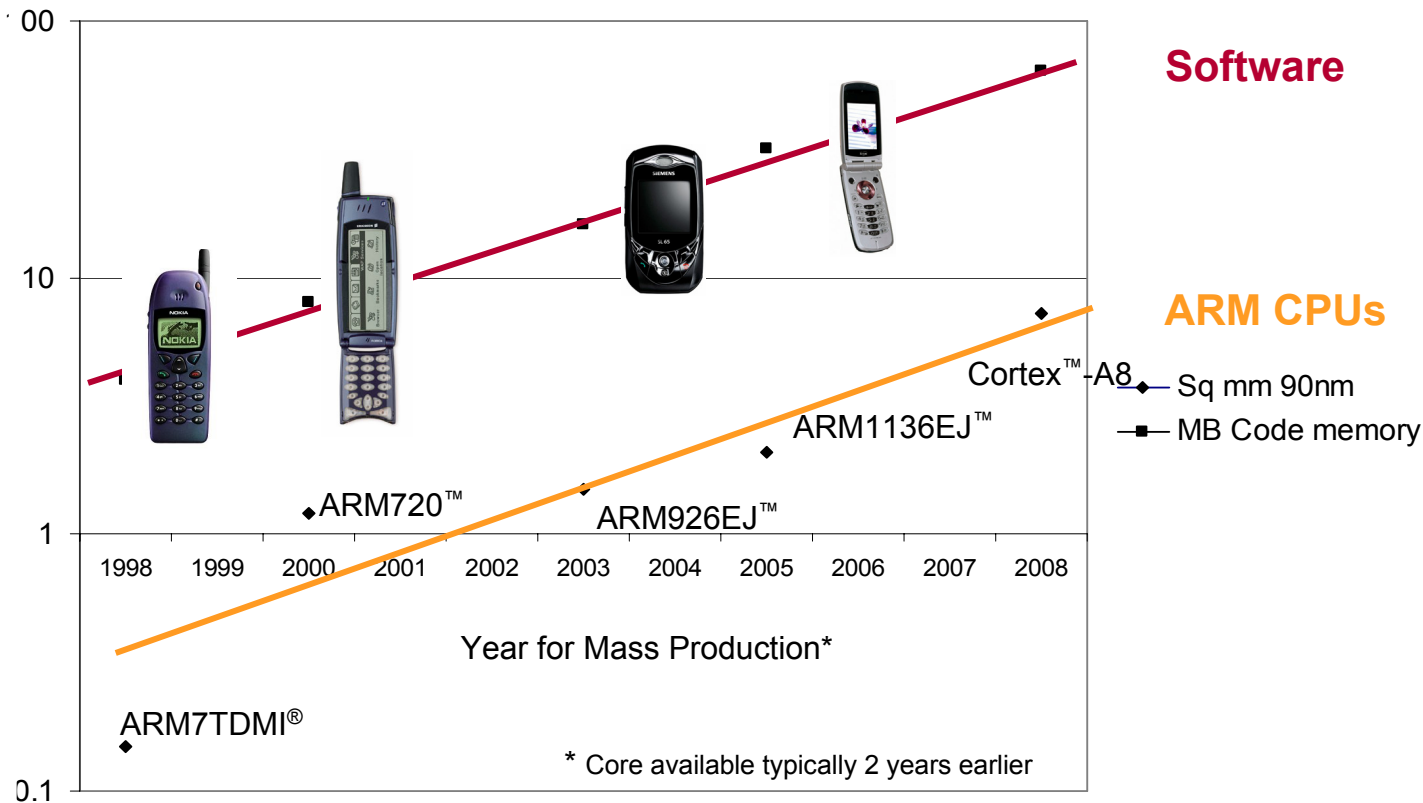
ARM9 family
100-250MHz
Portable Media Player (Software or AudioDE)
Good Java experience (Jazelle)
PS1 equiv gaming experience (Swerve client)
Adobe Flashlite UI
\$3 per game

ARM7 family
<50MHz
\$ free per game



Moore's Law and 1mm² ARM1176EJ™

- Processor complexity doubling every 2 years
- Process improvements keeps die size and cost down



ARM1176JZ-S
1sq mm on 65nm
(Metro™ libraries)

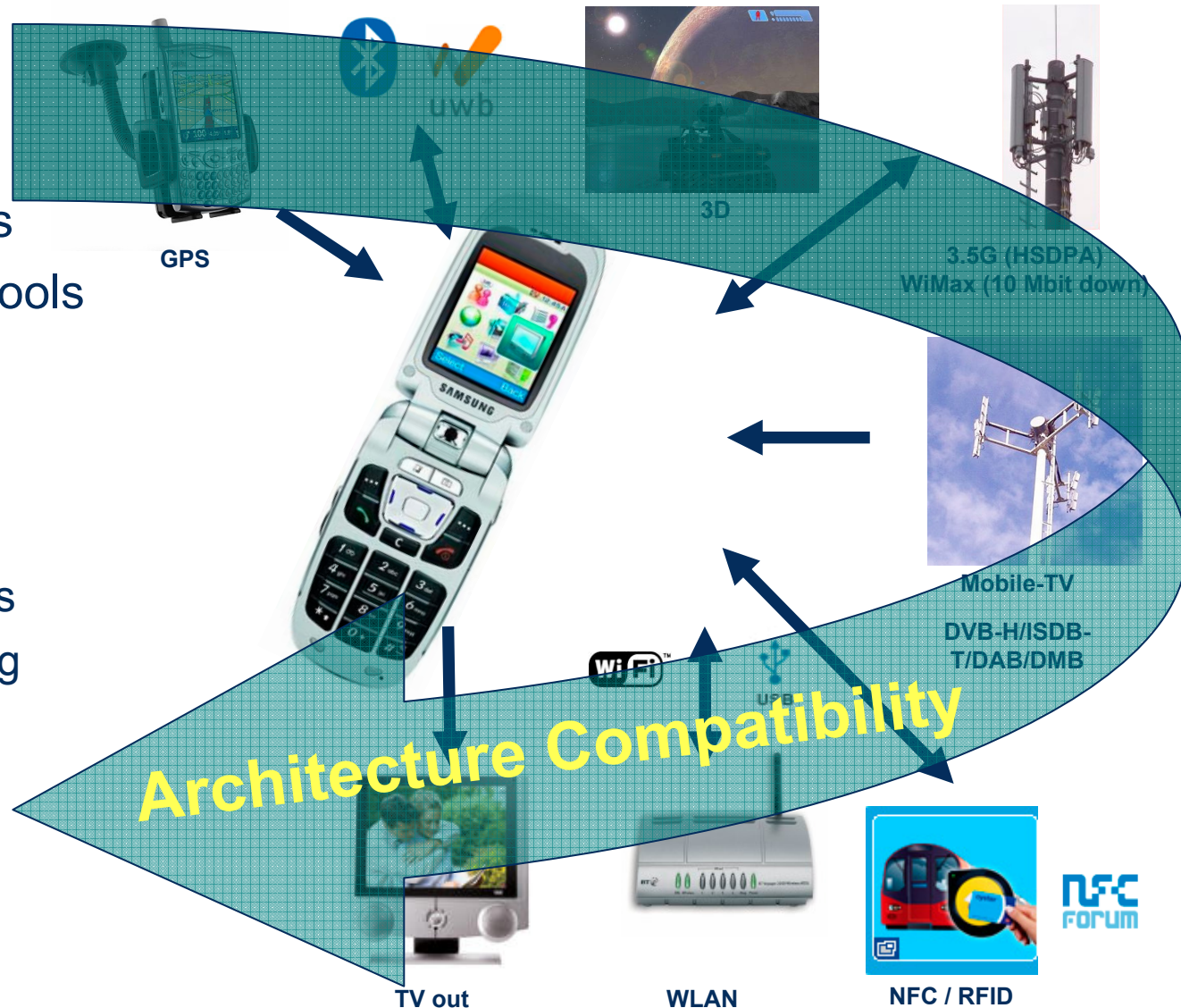
ARM Reduces Software Costs

- One architecture to optimise, one set of RealView® tools to create your product portfolio
- Increasing software complexity is driving operators towards Symbian/Windows Mobile/Linux
- PC applications are being optimized for ARM platform



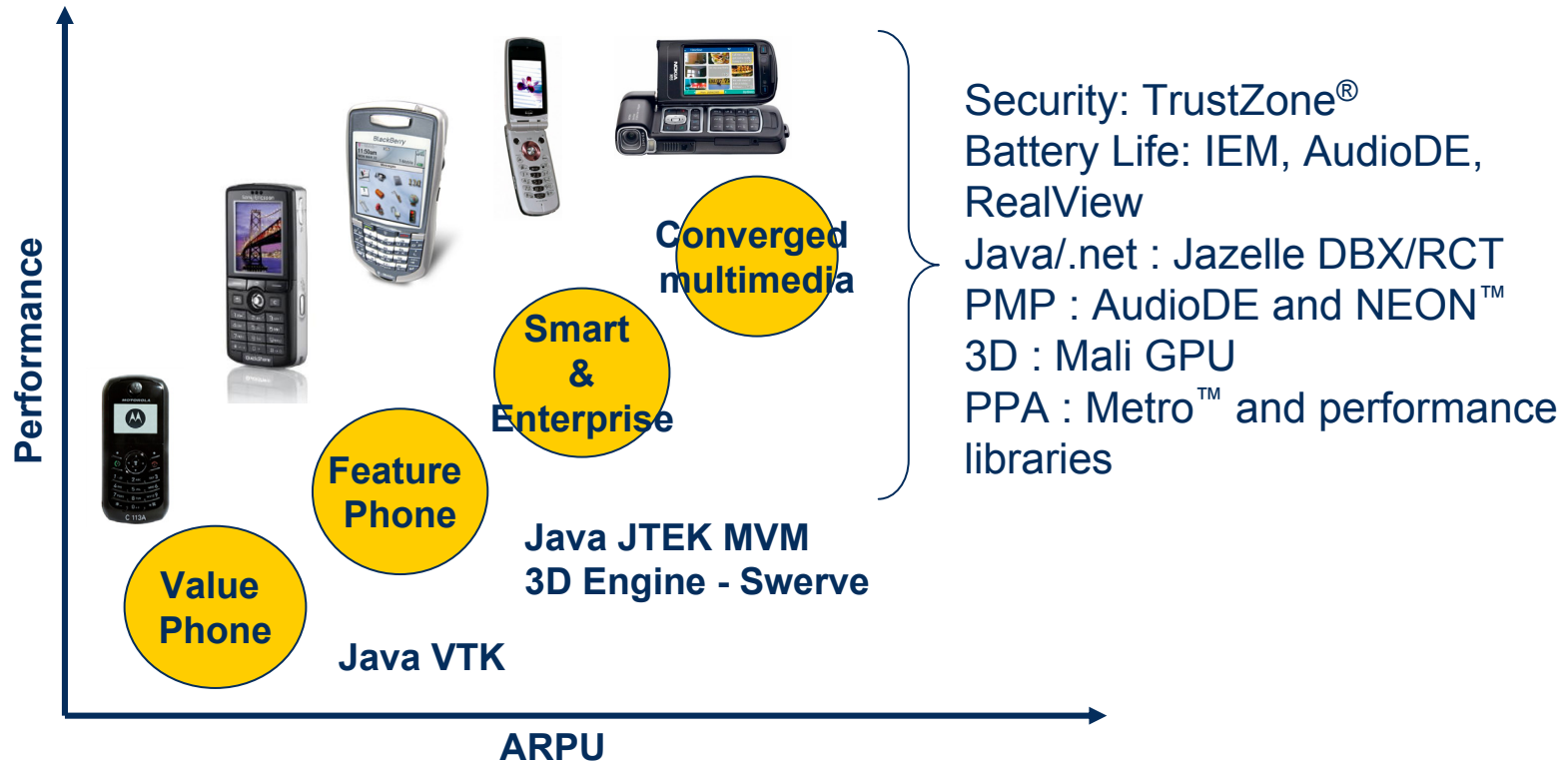
ARM Eases Hardware Integration

- Requires a mix of controllers, apps processors, data engines, peripherals
- System simulation tools and models
 - RealView SoC Designer
 - AMBA[®] Designer
 - AMBA peripherals
 - CoreSight[™] debug



Market Segmentation

- ARM has created the roadmap that spans the range:
 - \$20 Ultra Low Cost Handset (Cortex™-M3 or ARM7 processor)
 - \$300 Handheld Multimedia Computer (Cortex-A8 processor)



The Feature Phone Challenge

- The software challenge
 - Feature growth – ‘000s operator requirements?
 - Tight integration
 - System stability
- Wide range of software running
- Platform evolution
 - An ARM1176JZ-S™ processor comes with :
 - Double the performance
 - Predictive dynamic F & V scaling
 - TrustZone security framework
 - Latest AMBA 3 AXI protocol
 - 1sq mm on 65nm
 - Expect migration to ARM11 family by 2008

Voice

SMS

WAP

Ringtones

Camera

Java

Bluetooth

Audio Player

3D Graphics

QVGA Video

aGPS

3M+ Cameras

Picture Editing

Flash

MVM

It's Not Just Phones

■ Bluetooth

- Has become mass market
 - Specs keep evolving and rise of combo chips
 - Cortex-M3 processor, AudioDE, Metro libraries



■ Portable Media Players

- Gaming and video as well as audio
 - ARM1176JZ-S, AudioDE, Mali 3D, ESL, Metro libraries



■ Mobile GPS

- Stand alone units sales soar mass market in phones soon
 - Cortex-M3, ARM9, ARM11, Linux aware tools



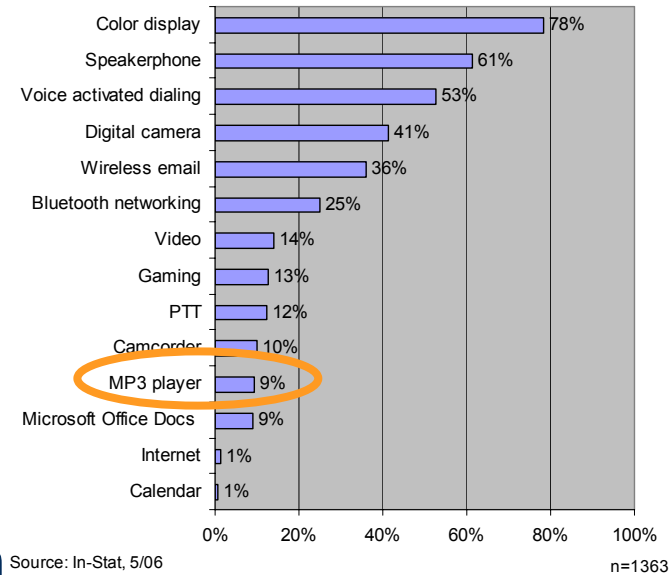
■ VOIP

- Pure voice over Wi-Fi as well as in cellular phones
 - ARM9, ARM11 family processors, AudioDE,

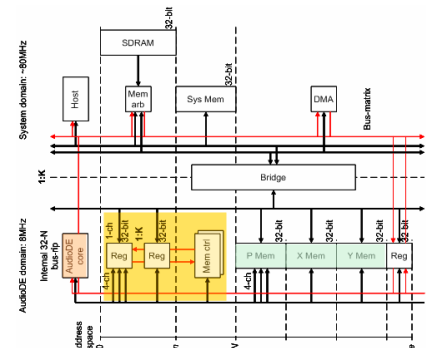


AudioDE – Play Music for Days

- If your PMP or music phone could play music for days would you use it more?
 - AudioDE delivers stunning benefits
 - (up to >40% in system)
 - Reprogrammable for multi CODEC
 - MP3, AAC, WMA, AMR etc
 - Ideal for PMP, Phones, VoIP, Bluetooth
 - 6.3MHz* measured playing a test track*
 - 0.65mW on 0.13um 1.2v TSMC
 - 20KBytes program memory
 - Minimized memory access
- *128Kbps MP3 bitstream, 44.1KHz, stereo output



Features used on a phone



**AudioDE Reference
subsystem
< 1.5mW & 50KG**

Key Messages

- ARM silicon Partners, OEMs and the entire value chain shipped 1 billion ARM processors into mobile devices in 2005
- The ARM roadmap for mobile is driven largely by:
 - Convergence and cost
 - Operators need to have secure capable platforms for services (\$)
 - Best in class power, performance, area and value
 - E.g. Cortex-M3, AudioDE, Metro/Advantage Libs, RealView
 - The best tools from modelling through design to debug
- Going up ARM's roadmap only costs cents but translates into \$\$ of added value for OEM's and operators
- ARM7 and ARM9 processor powered solutions have changed the world
- The high-performance roadmap redefines what is possible